

TEACHING NOTE

MARSHALLIAN DEMAND SYSTEMS FOR COMMON UTILITY FUNCTIONS

ECON 101–INTRODUCTION TO MICROECONOMICS

FALL 2023

PROF. ALEJANDRO MELO PONCE

NAZARBAYEV UNIVERSITY

These notes show how to derive the Marshallian demand system for a consumer arising from the **Utility Maximization Problem**:

$$\text{maximize}_{x_1, x_2} u(x_1, x_2) \text{ subject to } p_1 x_1 + p_2 x_2 \leq I,$$

where p_1 and p_2 are strictly positive prices and I is a strictly positive level of income.

The basic steps are:

1. Obtain the marginal utilities for good 1 and good 2. (Usually given as data for the problem.)
2. Obtain the marginal rate of substitution, where you need to decide which good goes in the horizontal axis and which good goes in the vertical axis. In these notes, I will always put good 1 in the horizontal axis and good 2 in the vertical axis.
3. Write the optimality condition for an interior optimum. (In the case you have a problem with an interior optimum and not a corner solution.)
4. From the optimality condition, solve for one of the variables.
5. Substitute what you obtained in the previous step into the budget constraint and solve for the optimal level of one of the goods as a function of the parameters (prices and income). This will be the Marshallian demand function for one of the goods.
6. With the Marshallian demand of one of the goods, use the expression you obtained in step 4 to solve for the other Marshallian demand. The two expressions you obtained now form the **Marshallian Demand System** for the consumer.

1 Cobb-Douglas Utility

$$u(x_1, x_2) = x_1^\alpha x_2^{1-\alpha}$$

where $0 < \alpha < 1$.

The marginal utilities are: $MU_1 = \alpha x_1^{\alpha-1} x_2^{1-\alpha}$ and $MU_2 = (1-\alpha) x_1^\alpha x_2^{-\alpha}$. Then, the marginal rate of substitution is

$$MRS = \frac{MU_1}{MU_2} = \frac{\alpha x_1^{\alpha-1} x_2^{1-\alpha}}{(1-\alpha) x_1^\alpha x_2^{-\alpha}} = \frac{\alpha x_2}{(1-\alpha) x_1}.$$

The optimality condition for an interior optimum is:

$$\frac{\alpha x_2}{(1-\alpha) x_1} = \frac{p_1}{p_2},$$

which implies that

$$x_2 = \frac{1-\alpha}{\alpha} \frac{p_1}{p_2} x_1.$$

Substituting the previous expression into the budget constraint yields:

$$p_1 x_1^m + p_2 \left(\frac{1-\alpha}{\alpha} \frac{p_1}{p_2} x_1^m \right) = I \Rightarrow p_1 x_1^m \left(1 + \frac{1-\alpha}{\alpha} \right) = I \Rightarrow p_1 x_1^m \left(\frac{1}{\alpha} \right) = I \Rightarrow x_1^m = \frac{\alpha I}{p_1}.$$

Using the results from the previous two displayed equations yields:

$$x_2^m = \frac{1-\alpha}{\alpha} \frac{p_1}{p_2} \frac{\alpha I}{p_1} = \frac{(1-\alpha)I}{p_2}$$

The Marshallian Demand System for the Cobb-Douglas utility function is then:

$$x_1^m(p_1, p_2, I) = \frac{\alpha I}{p_1}, \quad x_2^m(p_1, p_2, I) = \frac{(1-\alpha)I}{p_2}$$

Note that if we define the **expenditure share** of each good as:

$$s_1 = \frac{p_1 x_1}{I}, \quad s_2 = \frac{p_2 x_2}{I},$$

then the expenditure shares for the Cobb-Douglas Marshallian demand system are simply: $s_1 = \alpha$ and $s_2 = 1 - \alpha$.

2 Constant-Elasticity of Substitution (CES) Utility

$$u(x_1, x_2) = \frac{x_1^\delta}{\delta} + \frac{x_2^\delta}{\delta}$$

where $\delta \leq 1$ and $\delta \neq 0$.

The marginal utilities are: $MU_1 = x_1^{\delta-1}$ and $MU_2 = x_2^{\delta-1}$. Then, the marginal rate of substitution is:

$$MRS = \frac{MU_1}{MU_2} = \frac{x_1^{\delta-1}}{x_2^{\delta-1}} = \left(\frac{x_1}{x_2} \right)^{\delta-1}.$$

The optimality condition for an interior optimum is:

$$\left(\frac{x_1}{x_2} \right)^{\delta-1} = \frac{p_1}{p_2},$$

which implies that

$$\frac{x_1}{x_2} = \left(\frac{p_1}{p_2} \right)^{\frac{1}{\delta-1}}$$

From here, you can solve for either x_1 or x_2 and plug into the budget constraint to obtain the demands, but this requires some tedious algebra. Thus, I will show you here another a trick that you can use that also has a lot of economic meaning. Multiply both sides of the previous equation by the price ratio p_1/p_2 to get:

$$\frac{p_1 x_1}{p_2 x_2} = \frac{p_1}{p_2} \left(\frac{p_1}{p_2} \right)^{\frac{1}{\delta-1}} = \left(\frac{p_1}{p_2} \right)^{1+\frac{1}{\delta-1}} = \left(\frac{p_1}{p_2} \right)^{\frac{\delta}{\delta-1}}.$$

The previous equation implies that the ratio of expenditure shares $s_1/s_2 = \left(\frac{p_1 x_1/I}{p_2 x_2/I} \right)$ is equal to:

$$\frac{p_1 x_1}{p_2 x_2} = \frac{I}{I} \frac{p_1 x_1}{p_2 x_2} = \frac{s_1}{s_2} = \left(\frac{p_1}{p_2} \right)^{\frac{\delta}{\delta-1}},$$

and therefore:

$$s_1 = \left(\frac{p_1}{p_2} \right)^{\frac{\delta}{\delta-1}} s_2.$$

Now, the budget line in terms of expenditure shares is: $s_1 + s_2 = 1$. We can use the previous two equations to solve for the optimal expenditure shares:

$$\left(\frac{p_1}{p_2} \right)^{\frac{\delta}{\delta-1}} s_2 + s_2 = 1 \Rightarrow s_2 \left(1 + \left(\frac{p_1}{p_2} \right)^{\frac{\delta}{\delta-1}} \right) = 1 \Rightarrow s_2 = \left(1 + \left(\frac{p_1}{p_2} \right)^{\frac{\delta}{\delta-1}} \right)^{-1}$$

Plugging the previous expression into the equation relating the two expenditure shares yields, after some algebra:

$$s_1 = \left(1 + \left(\frac{p_2}{p_1} \right)^{\frac{\delta}{\delta-1}} \right)^{-1}.$$

At this point you may be wondering, where is the economics behind what I have done. Notice that the optimal expenditure shares for each good from the CES utility function depend only on the price ratio and the parameter δ . In contrast with the Cobb-Douglas utility, these budget shares do change with changes in the price ratio. However, the expressions remain relatively simple. With a little bit more work (which you may learn in future courses), these shares can be put into a form that can be used for empirical work. The CES specification is widely used in trade theory to analyze the effects of trade tariffs and quotas, since they change the relative price ratios of goods.

With the optimal expenditure shares, we can compute the Marshallian demands as follows:

$$p_1 x_1^m / I = s_1(p_1/p_2) \Rightarrow x_1^m = \frac{I}{p_1} s_1(p_1/p_2), \quad p_2 x_2^m / I = s_2(p_1/p_2) \Rightarrow x_2^m = \frac{I}{p_2} s_2(p_1/p_2).$$

The Marshallian demand system for the CES utility function is then:

$$x_1^m(p_1, p_2, I) = \frac{I}{p_1} s_1(p_1/p_2), \quad x_2^m(p_1, p_2, I) = \frac{I}{p_2} s_2(p_1/p_2)$$

3 Perfect Complements Utility

$$u(x_1, x_2) = \min \left(\frac{x_1}{\alpha}, \frac{x_2}{\beta} \right),$$

where $\alpha > 0$ and $\beta > 0$. Marginal utilities and the marginal rate of substitution are undefined for this utility function. However, we know that the indifference curve look like “right-angles”. The set of points that are the kinks in the indifference curve map is given by

$$\frac{x_1}{\alpha} = \frac{x_2}{\beta} \Rightarrow x_1 = \frac{\alpha}{\beta} x_2.$$

Substitute the previous expression into the budget constraint to obtain:

$$p_1 \left(\frac{\alpha}{\beta} x_2^m \right) + p_2 x_2^m = I \Rightarrow x_2^m \left(\frac{\alpha p_1 + \beta p_2}{\beta} \right) = I \Rightarrow x_2^m = \frac{\beta I}{\alpha p_1 + \beta p_2}.$$

Using the results from the two previous displayed equations yields:

$$x_1^m = \frac{\alpha}{\beta} \frac{\beta I}{\alpha p_1 + \beta p_2} = \frac{\alpha I}{\alpha p_1 + \beta p_2}.$$

The Marshallian Demand System for the Perfect Complements utility function is then:

$$x_1^m(p_1, p_2, I) = \frac{\alpha I}{\alpha p_1 + \beta p_2}, \quad x_2^m(p_1, p_2, I) = \frac{\beta I}{\alpha p_1 + \beta p_2}$$

4 Perfect Substitutes Utility

$$u(x_1, x_2) = \alpha x_1 + \beta x_2,$$

where $\alpha > 0$ and $\beta > 0$. The marginal utilities are $MU_1 = \alpha$ and $MU_2 = \beta$. Then, the marginal rate of substitution is

$$MRS = \frac{MU_1}{MU_2} = \frac{\alpha}{\beta}.$$

As discussed in class, with perfect substitutes utility we will have corner solutions, so you should not start by equating the price ratio to the marginal rate of substitution.

Recall the discussion in class: the consumer's whole budget should go to the good that provides the highest utility per unit of money spent. Thus, if $\frac{\alpha}{p_1} > \frac{\beta}{p_2}$, which is equivalent to $MRS = \frac{\alpha}{\beta} > \frac{p_1}{p_2}$, then all the budget should be used to consume good 1: $x_1^m = \frac{I}{p_1}$, $x_2^m = 0$. On the other hand, if $\frac{\alpha}{p_1} < \frac{\beta}{p_2}$, which is equivalent to $MRS = \frac{\alpha}{\beta} < \frac{p_1}{p_2}$, then all the budget should be used to consume good 2: $x_1^m = 0$, $x_2^m = \frac{I}{p_2}$. If it is the case that $\frac{\alpha}{p_1} = \frac{\beta}{p_2}$, which is equivalent to $MRS = \frac{\alpha}{\beta} = \frac{p_1}{p_2}$, then any point in the budget line is optimal: $x_1^m \in [0, I/p_1]$ and $x_2^m = I/p_2 - (p_1/p_2)x_1^m$.

$$x_1^m(p_1, p_2, I) = \begin{cases} \frac{I}{p_1} & \text{if } \frac{\alpha}{\beta} > \frac{p_1}{p_2} \\ 0 & \text{if } \frac{\alpha}{\beta} < \frac{p_1}{p_2} \\ [0, I/p_1] & \text{if } \frac{\alpha}{\beta} = \frac{p_1}{p_2} \end{cases}, \quad x_2^m(p_1, p_2, I) = \begin{cases} 0 & \text{if } \frac{\alpha}{\beta} > \frac{p_1}{p_2} \\ \frac{I}{p_2} & \text{if } \frac{\alpha}{\beta} < \frac{p_1}{p_2} \\ \frac{I}{p_2} - \left(\frac{p_1}{p_2}\right)x_1^m(p_1, p_2, I) & \text{if } \frac{\alpha}{\beta} = \frac{p_1}{p_2} \end{cases}$$